Partial Differential Equation

Introduction to Partial Differential Equation

Initial Value and Boundary Value

For ODE, it's not surprising that an n-order equation has infinite solutions, depending on n arbitrary constants. However, for PDE, we can expect that an n-order PDE with m independent variables has infinite solution depending on $m-1$ independent variables and n arbitrary functions. In order to select desired solutions, we need initial value or boundary value.

Boundary value conditions can be mainly divided into three categories:

- **Dirichlet boundary condition**: The value of the solution is given on boundary.
- **Neumann boundary condition**: Normal derivative of solution is given on boundary.
- **Mixed boundary condition**: Both kinds of boundary conditions are applied.

Types of PDE

For second-order linear PDE, it can be divided into three types: Elliptic, Parabolic, and Hyperbolic. The general form of 2rd-order linear PDE is

$$\sum_{i,j=1}^{n} a_{ij}(x)\frac{\partial^2 u}{\partial x_i \partial x_j} + (\text{lower order terms}) = 0$$

The coefficient matrix $A(x) = [a_{ij}(x)]$ is symmetric and its eigenvalues are used as the determinant.

- All $\lambda \neq 0$ and **same sign** $\implies$ Elliptic: steady-state or equilibrium
- All $\lambda \geq 0$ and **some = 0** $\implies$ Parabolic: diffusion-like processes or evolution toward equilibrium
- Exactly one $\lambda_i = 0$ $\implies$ Hyperbolic: wave-like or signal propagation
  For 2D PDE, the simplified determinant $\Delta = B^2 - AC$ can be used for

$$A\frac{\partial^2 u}{\partial x^2} + 2B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + \text{lower order terms} = 0$$

- $\Delta < 0 \implies$ Elliptic
- $\Delta = 0 \implies$ Parabolic
- $\Delta > 0 \implies$ Hyperbolic

Linear and non-linear wave

Stationary Wave, Travelling wave, and Transport

The simplest example of PDE about $u(x,t)$ may be:

$$\frac{\partial u}{\partial t}=0$$

By integral we have:

$$0=\int_0^t \frac{\partial u}{\partial s}(x,s)\mathrm{d}s=u(x,t)-u(x,0)$$

Therefore we have a classic solution: $u=u(x,0)$, which represents a stationary wave. This example may be too simple because x here is indeed a parameter rather than an independent variable. Consider the following example:

$$\frac{\partial u}{\partial t}+c\frac{\partial u}{\partial x}=0$$

where c is wave speed. We call this PDE as transport equation for it simulates the transportation of matters. We can import the characteristic variable $\xi=x-ct$ and $u(x,t)=v(x-ct,t)=v(\xi,t)$. we can rewrite the equation:

$$\frac{\partial u}{\partial t}+c\frac{\partial u}{\partial x}=\frac{\partial v}{\partial t}-c\frac{\partial v}{\partial \xi}+c\frac{\partial v}{\partial \xi}=\frac{\partial v}{\partial t}$$

Therefore $v(x,t)$ is a solution if and only if

$$\frac{\partial v}{\partial t}=0,$$

which means u must be a function of ξ. We can see that for this equation, signals propagate along characteristics. Consider another constant $a>0$,

$$\frac{\partial u}{\partial t}+c\frac{\partial u}{\partial x}+au=0$$

is called damped transport equation. Its solution is given by

$$u(x,t)=u(x-ct,0)e^{-at}$$

If the wave speed is not constant, then the wave will propagate along characteristic curve on the $xt-$plane. Say solution $x(t)$ to the autonomous ODE

$$\frac{\mathrm{d}x}{\mathrm{d}t}=c(x)$$

is the characteristic curve $\beta(x)$ of the transport function with wave speed $c(x)$. Then the solution is:

$$u(x,t) = u(\beta^{-1}(\beta(x) - t), 0)$$

Example: Solve the following equation:

$$\frac{\partial u}{\partial t} + \frac{1}{x^2+1}\frac{\partial u}{\partial x} = 0$$

We can solve the characteristic curve:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{x^2+1}$$

we can get $\beta(x) : \frac{1}{3}x^3 + x = t + C$. Therefore the characteristic variable is $\xi = \frac{1}{3}x^3 + x - t$ and the solution is

$$u = v\left(\frac{1}{3}x^3 + x - t\right)$$

where $v(x,0) = \frac{1}{1+(x+3)^2}$, is the initial value.

d'Alembert Equation

Govern equation of small vibration in one-dimensional medium:

$$\rho(x)\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x}\left(\kappa(x)\frac{\partial u}{\partial x}\right).$$

It's derived from Newton's 2nd law, and we can see the RHS is the restoring forces due to the small deformation. If the medium is uniform, we get the one-dimensional wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}, \quad \text{where the constant} \quad c = \sqrt{\frac{\kappa}{\rho}} > 0$$

If u satisfies $u_t + cu_x = 0$, it must be a solution to original PDE. Similarly, any backward wave such that $u_t - cu_x = 0$ is also a solution. Therefore, we know that this wave equation is bidirectional.

Theorem: Every solution to the wave equation can be written as a superposition of right and left traveling waves:

$$u(t,x) = p(\xi) + q(\eta) = p(x - ct) + q(x + ct),$$

where $p(\xi)$ and $q(\eta)$ are arbitrary function of characteristic variable. For the initial value:

$$u(0,x) = p(x) + q(x) = f(x), \quad \frac{\partial u}{\partial t}(0,x) = -cp'(x) + cq'(x) = g(x),$$

We have the following solution:

$$u(t,x) = \frac{f(x-ct)+f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(z)dz.$$

The first term represents two traveling waves moving in opposite directions without change in shape, and the integral term accounts for the initial velocity.

External Forcing and Resonance

When a homogeneous vibrating medium is subjected to external forcing term $F(x,t)$, the PDE will be

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2} + F(x,t)$$

with initial conditions:

$$u(x,0) = f(x), \quad \frac{\partial u}{\partial t}(x,0) = g(x)$$

The solution is given by

$$u(x,t) = u_h(x,t) + u_p(x,t)$$

where u_h is the homogeneous wave solution, and u_p is a **particular solution** that accounts for the forcing term. According to Duhamel's Principle, we know

$$u_p(x,t) = \frac{1}{2c}\int_0^t \int_{x-c(t-\tau)}^{x+c(t-\tau)} F(s,\tau)dsd\tau$$

Fourier Series

Half-Range Fourier Series

The basic form of Fourier series is given by

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos\left(\frac{n\pi x}{L}\right) + b_n\sin\left(\frac{n\pi x}{L}\right)\right]$$

where

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx, \quad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

These coefficients **project** f(x)f(x)f(x) onto the **orthogonal basis** of sine and cosine functions.

Suppose $f(x)$ is defined on $[0,L]$. There are two ways to extend it to $[-L,L]$:

- Even extension: reflect symmetrically $\rightarrow$ leads to a cosine series (Dirichlet BCs).
- Odd extension: reflect antisymmetrically $\rightarrow$ leads to a sine series (Neumann BCs).
  Thus, on $[0, L]$:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) \sim b_n \sin\left(\frac{n\pi x}{L}\right)$$

Fourier Transform

see integral transform.

Eigen solutions of Linear Evolution Equations

Separation of Variables and Eigenfunction Expansion

If the PDE is linear and homogeneous, (also BC is homogeneous) we assume the solution can be written as a product of temporal part and spatial part. For example, consider 1D wave equation with Dirichlet BC (fixed ends):

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < L, t > 0$$

and initial condition:

$$u(x, 0) = f(x), \quad \frac{\partial u}{\partial t}(x, 0) = g(x)$$

we can write the solution as:

$$u(x, t) = X(x)T(t)$$

Substitute into the wave equation

$$X(x)T''(t) = c^2 X''(x)T(t) \quad \Rightarrow \quad \frac{T''(t)}{c^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

This gives two ODEs, Spatial ODE:

$$X''(x) + \lambda X(x) = 0, \quad X(0) = X(L) = 0$$

and temporal ODE:

$$T''(t) + c^2 \lambda T(t) = 0$$

For spatial ODE, we know the eigenfunctions of Laplacian on $[0, L]$ with Dirichlet BC is $\left\{\sin\left(\frac{n\pi x}{L}\right)\right\}$, where $\lambda = \left(\frac{n\pi}{L}\right)^2, n = 1, 2, 3, \ldots$ For temporal ODE, we have

$T_n(t) = A_n \cos\left(\frac{n\pi ct}{L}\right) + B_n \sin\left(\frac{n\pi ct}{L}\right)$. The total solution is given by

$$u(x,t) = \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{n\pi ct}{L}\right) + B_n \sin\left(\frac{n\pi ct}{L}\right)\right] \sin\left(\frac{n\pi x}{L}\right)$$

Coefficients are determined by initial conditions

$$A_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)dx, \quad B_n = \frac{2}{n\pi c}\int_0^L g(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

Sturm–Liouville Theory

Sturm–Liouville theory is a cornerstone of solving PDEs by separation of variables. A regular Sturm–Liouville problem is a second-order ODE of the form:

$$\frac{d}{dx}\left[p(x)\frac{dy}{dx}\right] + [\lambda w(x) - q(x)]y = 0,$$

on $a < x < b$ with BC:

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0, \quad \beta_1 y(b) + \beta_2 y'(b) = 0$$

We can write the S–L equation as:

$$L[y] = -\frac{d}{dx}\left[p(x)\frac{dy}{dx}\right] + q(x)y = \lambda w(x)y$$

Here L is a **self-adjoint differential operator** under the inner product:

$$\langle f, g\rangle = \int_a^b f(x)g(x)w(x)dx$$

If p, p', q, w are continuous and $p, w > 0$ then the set $\{y_n(x)\}$ forms a complete basis in $L^2_w(a,b)$, meaning any reasonable function can be expanded as:

$$f(x) = \sum_{n=1}^{\infty} c_n y_n(x),$$

where

$$c_n = \frac{\int_a^b f(x)y_n(x)w(x)dx}{\int_a^b y_n(x)^2 w(x)dx}.$$

Special Functions

Green Functions method

Green’s function method is one of the most powerful tools for solving linear inhomogeneous PDEs, For a linear differential operator L acting on u:

$$Lu(x) = f(x), \quad x \in \Omega,$$

with given boundary conditions (BCs). The Green's function $G(x, \xi)$ is defined as the solution to:

`LG(x, \xi) = \delta(x - \xi), \quad \text{with the same BCs in } x,$$By analogy, here the`

\int{0}^{1} r J{\nu}(\alpha{\nu,m} r) J{\nu}(\alpha_{\nu,n} r) , dr = 0 \quad (m \neq n)$$

- Zeros: Important for eigenvalue problems with boundary conditions $R(a) = 0$.

Legendre Function

Legendre functions arise when solving PDEs in spherical coordinates. For example, Laplace's equation in spherical coordinates (r, θ, ϕ):

$$\frac{\partial}{\partial r}\left(r^2 \frac{\partial u}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial u}{\partial \theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 u}{\partial \phi^2} = 0$$

Separating θ and ϕ gives the associated Legendre equation. Legendre equation:

$$(1 - x^2)y'' - 2xy' + \ell(\ell+1)y = 0$$

Properties:

- Orthogonality:

$$\int_{-1}^{1} P_\ell(x)P_m(x)\,dx = \frac{2}{2\ell+1}\delta_{\ell m}$$

- Normalization: $P_\ell(1) = 1$
- Recurrence:

$$(\ell+1)P_{\ell+1}(x) = (2\ell+1)xP_\ell(x) - \ell P_{\ell-1}(x)$$

When separation in ϕ gives a factor $e^{im\phi}$, the θ-part satisfies:

$$(1 - x^2)y'' - 2xy' + \left[\ell(\ell+1) - \frac{m^2}{1-x^2}\right]y = 0,$$

with $x = \cos\theta$. Solutions are associated Legendre functions:

$$P_\ell^m(x) = (-1)^m(1-x^2)^{m/2}\frac{d^m}{dx^m}P_\ell(x)$$